Q1 - 2024 (04 Apr Shift 1)

Let A be a square matrix of order 2 such that $|A|=2$ and the sum of its diagonal elements is -3 . If the points (x, y) satisfying $\mathrm{A}^2+x \mathrm{~A}+y \mathrm{I}=\mathrm{O}$ lie on a hyperbola, whose length of semi major axis is x and semi minor axis is y, eccentricity is e and the length of the latus rectum is l, then $81\left(e^4+l^2\right)$ is equal to

Q2 - 2024 (04 Apr Shift 2)

Consider a hyperbola H having centre at the origin and foci on the x-axis. Let C_1 be the circle touching the hyperbola H and having the centre at the origin. Let C_2 be the circle touching the hyperbola H at its vertex and having the centre at one of its foci. If areas (in sq units) of C_1 and C_2 are 36π and 4π, respectively, then the length (in units) of latus rectum of H is

(1) $\frac{14}{3}$
(2) $\frac{28}{3}$
(3) $\frac{11}{3}$
(4) $\frac{10}{3}$

Q3 - 2024 (06 Apr Shift 2)

The length of the latus rectum and directrices of a hyperbola with eccentricity e are 9 and $x=\pm\frac{4}{\sqrt{13}}$, respectively. Let the line $y-\sqrt{3}x+\sqrt{3}=0$ touch this hyperbola at (x_0, y_0). If m is the product of the focal distances of the point (x_0, y_0), then $4\mathrm{e}^2+\mathrm{m}$ is equal to ________

Q4 - 2024 (08 Apr Shift 1)

Let $H: \frac{-x^2}{a^2}+\frac{y^2}{b^2}=1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus rectum is $4\sqrt{3}$. Suppose the point $(\alpha, 6), \alpha>0$ lies on H. If β is the product of the focal distances of the point $(\alpha, 6)$, then $\alpha^2+\beta$ is equal to

(1) 172

(2) 171

(3) 169

(4) 170

Q5 - 2024 (08 Apr Shift 2)

Let S be the focus of the hyperbola $\frac{x^2}{3}-\frac{y^2}{5}=1$, on the positive x-axis. Let C be the circle with its centre at $A(\sqrt{6}, \sqrt{5})$ and passing through the point S. If O is the origin and SAB is a diameter of C, then the square of the area of the triangle OSB is equal to____________

Q6 - 2024 (09 Apr Shift 2)

Let the foci of a hyperbola H coincide with the foci of the ellipse $E: \frac{(x-1)^2}{100}+\frac{(y-1)^2}{75}=1$ and the eccentricity of the hyperbola H be the reciprocal of the eccentricity of the ellipse E. If the length of the transverse axis of H is α and the length of its conjugate axis is β, then $3\alpha^2+2\beta^2$ is equal to

(1) 237

(2) 242

(3) 205

(4) 225

Answer Key

Q1 (233) **Q2** (2) **Q3** (61) **Q4** (2)

Q5 (40) **Q6** (4)

Q1

Given $|A| = 2$

trace $A = -3$

and $A^2 + xA + yI = 0$

$\Rightarrow x = 3, y = 2$

so, $81e^4 = 169$

$LR = 8/9$

$81e^4 + 81L^2 = 169 + 64 = 233$

Q2

Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \left(b^2 = a^2\left(e^2 - 1\right)\right)$

$\therefore$ eq^n of C_1

Ar. $= 36\pi$

$\pi a^2 = 36\pi$

$a = 6$

$\therefore$ eq^n of $C_1 = x^2 + y^2 = a^2$

Now radius of C_2 can be $a(e-1)$ or $a(e+1)$ for $r = a(e-1)$ for $r = a(e+1)$

Ar. $= 4\pi \; \pi r^2 = 4\pi$

$\pi a^2(e-1)^2 = 4\pi$

$a^2(e+1)^2 = 4$

$36\pi(e-1)^2 = 4\pi$

$36(e+1)^2 = 4$

$e - 1 = \frac{1}{3}$

$e + 1 = \frac{1}{3}$

$e = \frac{4}{3}$

$-\frac{2}{3}$

Not possible

$\therefore b^2 = 36\left(\frac{16}{9} - 1\right) = 28$

$\therefore LR = \frac{2\,b^2}{a} = \frac{2 \times 28}{6} = \frac{28}{3}$

Q3

Given $\frac{2\mathrm{~b}^{2}}{\mathrm{a}}=9$ and $\frac{\mathrm{a}}{\mathrm{e}}=\pm\frac{4}{\sqrt{3}}$

equation of tangent $y-\sqrt{3}x+\sqrt{3}=0$

by equation of tangent

Let slope $=\mathrm{S}=\sqrt{3}$

Constant $=-\sqrt{3}$

By condition of tangency

$$\Rightarrow 6=6\mathrm{a}^{2}-9\mathrm{a}$$
$$\Rightarrow \mathrm{a}=2,\ \mathrm{b}^{2}=9$$

Equation of Hyperbola is

$\frac{x^{2}}{4}-\frac{y^{2}}{9}=1$ and for tangent

Point of contact is $(4,3\sqrt{3})=(\mathrm{x}_{0},\mathrm{y}_{0})$

Now $\mathrm{e}=\sqrt{1+\frac{9}{4}}=\frac{\sqrt{13}}{2}$

Again product of focal distances

$$\mathrm{m}=(\mathrm{x}_{0}\mathrm{e}+\mathrm{a})(\mathrm{x}_{0}\mathrm{e}-\mathrm{a})$$
$$\mathrm{m}+4\mathrm{e}^{2}=20\mathrm{e}^{2}-\mathrm{a}^{2}$$
$$=20\times\frac{13}{4}-4=61$$

(There is a printing mistake in the equation of directrix $x=\pm\frac{4}{\sqrt{3}}$.

Corrected equation is $x=\pm\frac{4}{\sqrt{13}}$ for directrix, as eccentricity must be greater than one, so question must be bonus)

Q4

$$\mathrm{H}:\frac{\mathrm{y}^{2}}{\mathrm{b}^{2}}-\frac{\mathrm{x}^{2}}{\mathrm{a}^{2}}=1,\mathrm{e}=\sqrt{3}$$
$$\mathrm{e}=\sqrt{1+\frac{\mathrm{a}^{2}}{\mathrm{b}^{2}}}=\sqrt{3}\Rightarrow\frac{\mathrm{a}^{2}}{\mathrm{b}^{2}}=2$$
$$\mathrm{a}^{2}=2\mathrm{~b}^{2}$$
$$\text{length of L.R.}=\frac{2\mathrm{a}^{2}}{\mathrm{b}}=4\sqrt{3}$$
$$a=\sqrt{6}$$

$P(\alpha,6)$ lie on $\frac{y^{2}}{3}-\frac{x^{2}}{6}=1$

$12-\frac{\alpha^2}{6}=1 \Rightarrow \alpha^2=66$

$\text{Foci }=(0, \pm \text{ be })=(0,3) \&(0,-3)$

Let $d_1 \& d_2$ be focal distances of $\mathrm{P}(\alpha, 6)$

$\mathrm{d}_1=\sqrt{\alpha^2+(6+b e)^2},\ \mathrm{d}_2=\sqrt{\alpha^2+(6-b e)^2}$

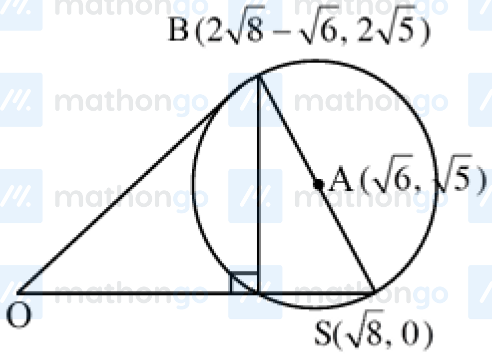
$\mathrm{d}_1=\sqrt{66+81},\ \mathrm{d}_2=\sqrt{66+9}$

$\beta=\mathrm{d}_1\ \mathrm{d}_2=\sqrt{147 \times 75}=105$

$\alpha^2+\beta=66+105=171$

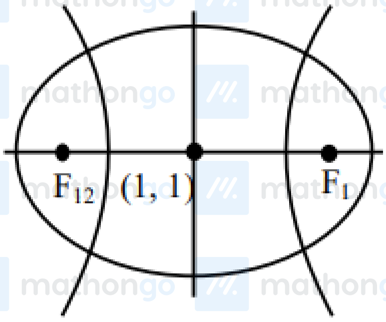
SECTION - B

Q5



Area $=\frac{1}{2}(O S) h=\frac{1}{2} \sqrt{8} 2 \sqrt{5}=\sqrt{40}$

Q6



$$e_1 = \sqrt{1 - \frac{75}{100}} = \frac{5}{10} = \frac{1}{2}$$

$e_2 = 2$

$F_1(6, 1), F_2(-4, 1)$

$2ae_2 = 10 \Rightarrow a = \frac{5}{2} \Rightarrow 2a = 5$

$\Rightarrow \alpha = 5$

$4 = 1 + \frac{b^2}{a^2} \Rightarrow b^2 = 3a^2$

$b = \sqrt{3} \times \frac{5}{2}$

$\beta = 5\sqrt{3}$

$3\alpha^2 + 2\beta^2 = 3 \times 25 + 2 \times 25 \times 3$

$= 225$